Firmware

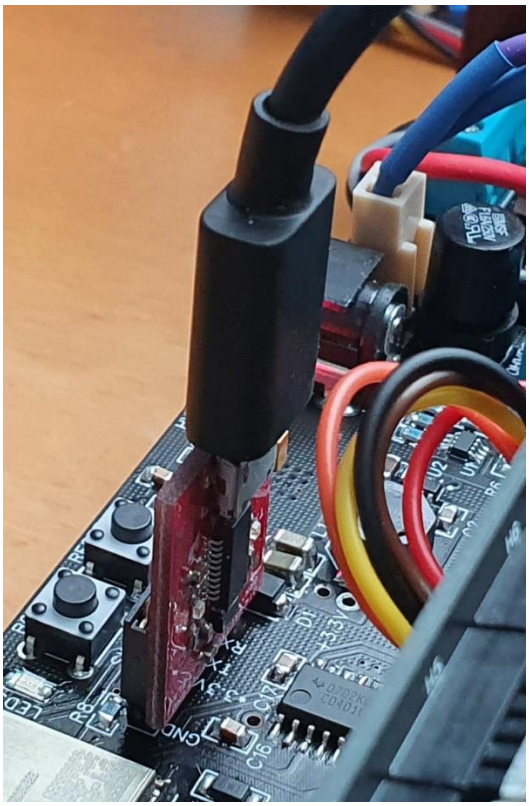
I use Visual Studio Code with the PlatformIO extension to edit, compile and flash the firmware.

To flash the firmware, you need a **3.3V** FTDI adapter with the standard pin-out.

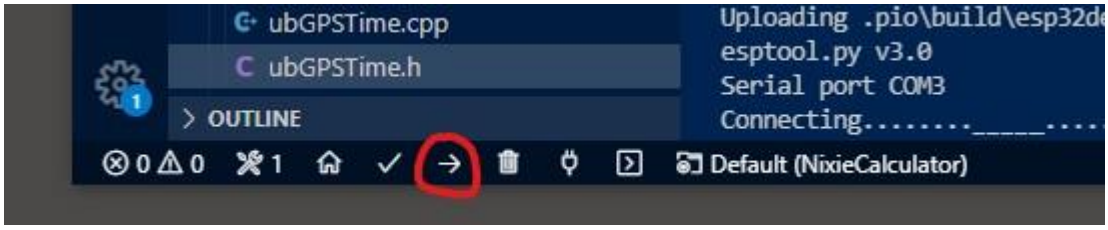
Flashing the controller firmware

Project folder: Software/Firmware/Controller

Power off the calculator and connect the FTDI adapter to the controller board. Make sure you connect your adapter the correct way around by checking the ground pin (GND).



To compile and flash the controller firmware, press the → symbol. Once you get the "Connecting…" message, press and hold the "boot" button on the controller board, then press and release the "reset" button and finally release the "boot" button. This should start the flash process.



Flashing the keyboard firmware

Project folder: Software/Firmware/Keyboard

Power off the calculator and connect the FTDI adapter to the keyboard. Make sure you connect your adapter the correct way around by checking the ground pin (GND).

To compile and flash the keyboard firmware just press the → symbol. The flash process should start automatically.

Starting the calculator

After flashing the controller and the keyboard firmware, **remove the FTDI adapter** and power on the calculator. During start-up, the calculator briefly shows the controller firmware version on the left and the keyboard firmware version on the right.